
AS-Regularized Deep Market Making without Hybrid Reward Shaping

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Abstract

Deep reinforcement-learning market makers are often trained with hybrid rewards that mix dampened profit-and-loss, transaction advantage, and inventory penalties. These terms work, but they bury the designer’s intent inside indistinct scalar coefficients that interact indirectly and must be re-tuned per market. We revisit the Attn-LOB / C-PPO market maker of Guo et al. [8] and ask whether the behavioural prior can instead come from a classical quoting model. We propose SOFT-AS PPO: PPO with pure profit as the economic reward and a soft action-space regularizer toward a per-stock Avellaneda–Stoikov (AS) teacher fitted on training data. This is a control-prior regularizer, not a reward heuristic: it still shapes the objective, but through an interpretable economic model whose parameters can be calibrated, logged, and stress-tested. On a synthetic, paper-shaped replication benchmark, SOFT-AS PPO wins 93% of 15 paired stock–seed runs against both controls (paired $\Delta\text{PnL} = 15.10$ over C-PPO, 95% CI [9.19, 21.01], $p = 8.1 \times 10^{-5}$; $\Delta\text{PnL} = 9.50$ over PROFIT PPO, $p = 1.9 \times 10^{-4}$), corresponding to a $\sim 50\%$ and $\sim 21\%$ gain in spread-normalized PnL. Because the prior lives in the action space we can read it directly: the policy’s mean action gap to the teacher falls from 0.468 to 0.230, showing that the regularizer moves quoting behavior, not just terminal PnL. The gain is not free, SOFT-AS PPO quotes tighter, fills more, carries more inventory, and degrades faster under high latency. On this benchmark, an AS control prior is a cleaner and more inspectable way to steer a deep market maker than hybrid reward shaping.

1 Introduction

Market making is the problem of continuously quoting bid and ask limit orders to earn the spread while managing inventory risk [18, 13]. A modern limit order book (LOB) is high-dimensional, noisy, and fast-moving, so recent work uses deep reinforcement learning (RL) to learn quoting policies directly from LOB state. The Attn-LOB / C-PPO agent of Guo et al. [8] is a strong example: a convolutional-attention encoder reads a rolling LOB window and PPO emits continuous bid/ask quote parameters.

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The agent’s behaviour, however, is largely determined by a hand-designed reward. The original C-PPO reward adds three heuristic terms: dampened PnL, trading PnL, and an inventory penalty:

$$R_t = \underbrace{\Delta \text{PnL}_t - \max(0, \eta \Delta \text{PnL}_t)}_{\text{dampened PnL}} + \underbrace{X_{t,v}(p_{m,t} - X_{t,p})}_{\text{trading PnL}} - \underbrace{\zeta \text{Inv}_t^2}_{\text{inventory penalty}}. \quad (1)$$

Each term encodes a sensible intention, e.g. suppress holding profits, reward fills near mid, discourage large positions, but the encoding is opaque. The terms are not potential-based [11], so they reshape the optimal policy rather than merely accelerating learning; their coefficients (η, ζ) interact indirectly, must be re-tuned per market, and give the designer no direct read on the quoting behaviour they induce. When such an agent quotes too tightly or carries too much inventory, the reward offers no interpretable knob with which to correct it.

We ask whether the behavioural prior can instead come from a calibrated classical model, expressed where it is easiest to inspect: the action itself. We keep the marked-to-market profit as the economic reward and add a soft regularizer that pulls the learned continuous action toward a fitted Avellaneda–Stoikov (AS) teacher [1]:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_t \Delta \text{PnL}_t - \lambda \|a_t^\pi - a_t^{\text{AS}}\|_2^2 \right]. \quad (2)$$

The AS term is part of the training objective, but the economic reward being optimized is ordinary marked-to-market profit. We do not present this as a way to escape objective design; it is a soft control-prior regularizer, a domain instance of pulling a deep policy toward a fixed policy prior in action space [2], and it changes what the agent optimizes just as the hybrid reward does. The difference is where the design lives. Rather than spreading the designer’s intent across reward coefficients that interact indirectly, we concentrate it in a single object: an Avellaneda–Stoikov teacher whose parameters γ and κ carry economic meaning and can be calibrated per stock, with one weight λ setting how hard the policy is held to it. Because the policy and the teacher emit actions in the same normalized space, we can also measure the gap between them directly, and so check whether the prior is actually shaping quotes rather than reading that off terminal PnL.

Contributions.

- We formulate AS guidance as a soft control-prior regularizer for continuous-action PPO market making, replacing the hand-crafted hybrid reward of Eq. (1) with a profit objective plus an interpretable, calibrated action prior.
- We calibrate one AS teacher per stock from the training window, getting a behavioural prior that is inspectable, loggable, and steerable through economically meaningful knobs (γ, κ, λ).
- On a matched synthetic replication of Guo et al. [8] with five seeds per stock, we show that SOFT-AS PPO improves spread-normalized PnL and Sharpe over both reward-shaped C-PPO and a profit-only control, and we add teacher-divergence, latency, and decision-trace diagnostics that show the prior steers quoting behaviour directly and locate where it breaks.

2 Related work

Classical market-making models derive quotes by balancing spread capture and inventory risk. In the AS model, with inventory q , mid-price s , volatility σ , risk aversion γ , liquidity κ , and horizon T , the reservation price and total spread are

$$r(s, q, t) = s - q\gamma\sigma^2(T - t), \quad \delta^a + \delta^b = \gamma\sigma^2(T - t) + \frac{2}{\gamma} \log\left(1 + \frac{\gamma}{\kappa}\right). \quad (3)$$

This structure is interpretable but depends on assumptions about price and arrival processes [7]. Deep LOB models such as DeepLOB and related architectures learn representations directly from order-book tensors [20, 19, 12], and RL market makers have used handcrafted state, discrete actions, and model-free policies [10, 17, 21, 14, 5]. Guo et al. [8] combine an Attn-LOB encoder with a continuous, AS-like action space and the hybrid reward in Eq. (1), trained with PPO [15]; we keep their encoder, simulator, action space, and training setup, and change only how the behavioural prior enters.

Regularizing a learned policy toward a prior. Keeping the profit reward while penalizing departures from a fixed action-space prior is a standard device in reinforcement learning. The closest precedent is Cheng et al. [2], who regularize a deep policy toward a control prior using a weight λ that is adapted over training, and show that the regularized policy approximately preserves the prior’s control-theoretic stability guarantees while reducing the variance of the policy gradient. Our penalty admits the same bias–variance reading: the AS teacher supplies structure where the data are uninformative, and λ governs how far the policy may move away from it. Comparable constraints appear in offline RL, where the policy is tied to a behaviour prior to curb value over-estimation [4], and in residual policy learning, where a learned term is added on top of a fixed controller [16]. What is new here is not the regularizer, which is shared across these settings, but the prior it pulls toward: a per-stock Avellaneda–Stoikov model whose parameters γ and κ are economically meaningful and can be calibrated and stress-tested, in place of a generic KL or proximity term.

Coupling AS with learning. Combining AS with RL is not itself new; what differs is the direction of the coupling. Falces Marín et al. [3] have a network set AS’s risk-aversion parameter and let AS produce the quotes, so learning tunes the classical model from within. We reverse this: the policy quotes directly, and AS enters only as a target it may leave when the data warrant. The classical model therefore supplies a behavioural prior rather than the quoting rule.

3 The replicated agent

We build on the Attn-LOB / C-PPO market maker of Guo et al. [8], which has four parts: an encoder that reads the LOB, an action space that turns network outputs into quotes, a reward that defines good market making, and a replay simulator for training and evaluation. We keep all four and change only how the behavioural prior is supplied; full reproduction details are in Appendix A.

State and encoder. The agent observes a rolling window of $T = 50$ LOB snapshots, each a 40-dimensional vector of the top-ten ask and bid prices and volumes. The Attn-LOB encoder reads this 50×40 tensor with convolutional filters, an Inception block, and a multi-head self-attention layer: the convolutions pick up spatial patterns across price levels within each snapshot, while the Inception and attention layers capture temporal dependencies across the window at multiple scales. Dynamic agent state (inventory, remaining time, and cash or marked-to-market value) is concatenated to the encoded features. We pre-train the encoder on a three-way mid-price direction task and reuse it as the PPO feature extractor, as in the original work.

Continuous action space. The policy outputs two actions $A_1, A_2 \in [0, 1]$. A_1 sets a reservation-price bias and A_2 the quoted spread:

$$\delta = A_1 \cdot \text{max_bias}, \quad p_r = p_m - \text{sign}(\text{Inv}) \delta, \quad \text{spread} = A_2 \cdot \text{max_spread}, \quad p_{a,b} = p_r \pm \frac{\text{spread}}{2}. \quad (4)$$

The agent thus quotes symmetrically around a reservation price p_r that it skews against its own inventory. This is deliberately the same shape as the AS closed form of Eq. (3): A_1 takes the role of the inventory-dependent reservation shift $q\gamma\sigma^2(T-t)$, and A_2 the role of the optimal spread $\delta^a + \delta^b$.

Both the policy and the AS model write into this same (A_1, A_2) space, which is what lets AS serve as a regularization target in Section 4.

Reward (baseline). Our primary point of comparison, C-PPO, trains PPO on the hybrid reward of Eq. (1) with the original coefficients $\eta = 0.5$ and $\zeta = 0.01$. As noted in the introduction, these terms are not potential-based [11] and reshape the optimal policy rather than merely accelerating learning. We carry the coefficients over unchanged rather than re-tuning them for this benchmark. This keeps the baseline faithful to the source, and the difficulty of re-tuning such coupled terms for each new market is itself part of what motivates moving the prior out of the reward.

Simulator. A replay simulator fills the agent’s quotes only when the recorded stream trades through the quoted price. There is no market impact and no transaction cost; the agent may run long or short and is liquidated at the end of each 2000-event episode. The simulator is identical for every method, so within-benchmark comparisons are fair. It is not, however, neutral across quoting styles: with no spread to pay and no impact to absorb, a frictionless tape rewards tight,

high-participation quoting and so flatters more aggressive policies. Absolute PnL should therefore be read as a benchmark quantity rather than a live-trading estimate, and the cross-method ranking should be weighed alongside the inventory and latency costs we report in Section 6. This mirrors the small-order assumption of the original work.

4 AS-regularized PPO

We change only the training objective of the agent in Section 3: it still optimizes marked-to-market profit, but we add a soft penalty pulling each quoting action toward a fitted Avellaneda–Stoikov teacher. Concretely, the per-step reward is

$$r_t = \Delta \text{PnL}_t - \lambda \|a_t^\pi - a_t^{\text{AS}}\|_2^2, \quad (5)$$

and PPO maximizes the resulting return $J(\pi) = \mathbb{E}_\pi[\sum_t r_t]$, which is Eq. (2). The penalty acts on the action sampled during rollout collection, so it is part of the reward the policy optimizes rather than a separate supervised loss on the network. This is what we mean by a control-prior regularizer: it modifies the objective, like the hybrid reward, but through a single interpretable target instead of several hand-set coefficients.

The AS teacher. The teacher converts the AS closed form of Eq. (3) into the same normalized (A_1, A_2) coordinates the policy uses in Eq. (4). Given inventory q , a volatility estimate $\hat{\sigma}_t$, and time-to-close $T - t$,

$$a_t^{\text{AS}} = \left(\text{clip}\left(\frac{|q| \gamma \hat{\sigma}_t^2 (T-t)}{\text{max_bias}}, 0, 1\right), \text{clip}\left(\frac{\delta^a + \delta^b}{\text{max_spread}}, 0, 1\right) \right), \quad \delta^a + \delta^b = \gamma \hat{\sigma}_t^2 (T-t) + \frac{2}{\gamma} \log\left(1 + \frac{\gamma}{\kappa}\right). \quad (6)$$

The first coordinate matches the inventory-dependent reservation shift of Eq. (4) and the second the optimal spread; clipping into $[0, 1]$ keeps the teacher inside the policy’s action range. The sign of the bias is supplied by $\text{sign}(\text{Inv})$ in Eq. (4), so the teacher carries only its magnitude $|q|$. Once (γ, κ) are fixed and $\hat{\sigma}_t$ is read from the state, the teacher is a deterministic function of the agent’s situation and adds no learnable parameters.

Calibration. We fit one teacher per stock from the training window only. The liquidity parameter κ is estimated from the mean level-one half-spread, giving $\kappa = 125.2, 23.9, 83.8$ for stocks 000001, 000858, and 002415. For risk aversion we use a fixed low-risk value $\gamma = 0.025$ for the reported policy; a volatility-fitted γ and a higher-risk teacher are evaluated as controls in Appendix B. The volatility term $\hat{\sigma}_t$ is a rolling realized-volatility estimate computed from the same LOB stream the policy sees. We freeze the calibration used at training time and reuse it at evaluation, rather than recalibrating on the test split, so the teacher cannot leak test-window information into the policy’s prior.

Methods compared. We report three PPO agents that share this state, encoder, action space, and training budget, and differ only in the objective. C-PPO is the reward-shaped baseline of Section 3. PROFIT PPO drops the hybrid terms and trains on profit alone ($\lambda = 0$ in Eq. (5)), isolating the effect of removing the old reward. SOFT-AS PPO adds the AS regularizer with $\lambda = 0.10$ and the low-risk per-stock teacher. The three thus form a clean decomposition: C-PPO $\rightarrow$ PROFIT PPO measures the effect of replacing the hand-crafted reward with pure profit, and PROFIT PPO $\rightarrow$ SOFT-AS PPO measures what the calibrated prior adds on top. Hard AS constraints, behavioural-cloning warm starts, a direct-AS teacher policy, and the λ sweep are appendix-only controls (Appendix B).

5 Experimental setup

A synthetic benchmark. The original Shenzhen Stock Exchange LOB data are proprietary, so we use a synthetic event-level LOB generator that preserves the experimental shape of Guo et al. [8]: three stock identities, top-ten book levels, stable intraday windows, ten training days, eleven test days, and 2000-event episodes. Each instrument has its own base price and generated training/test days; a latent fair value moves through stable, trending, and volatile regimes while the generator builds the surrounding book, trade stream, and market-order flow. The simulator replays this stream

and fills the agent’s quotes only when recorded trades cross the quote. There is no market impact and no transaction cost. The benchmark is useful for controlled comparisons and debugging because every method faces the same synthetic tapes, but it remains a proxy for the proprietary data used in the original paper. Appendix A gives the generator details.

Training protocol. Following the original code, we train one policy per stock rather than a single shared cross-stock policy. All three agents (Section 4) use the same state, encoder, action space, trade granularity, train/test split, and 400k PPO steps, and differ only in the objective. The main comparison runs five PPO seeds per stock for each of C-PPO, PROFIT PPO, and SOFT-AS PPO, giving 15 stock–seed units per method. The regularizer weight λ and the per-stock teacher were selected in an earlier sweep (Appendix B); the final comparison reports five seeds per stock for the selected setting.

Metrics. Our primary metric is spread-normalized PnL, $\text{ND-PnL} = \text{PnL}/(\bar{s} + 10^{-7})$, where $\text{PnL} = V_T - V_0$ is the change in marked-to-market value over an episode and $\bar{s}$ is the mean quoted spread. Dividing by $\bar{s}$ makes performance comparable across stocks whose price levels, and therefore absolute spreads and PnL differ substantially; it counts roughly how many spreads the agent captured. We also report raw PnL, Sharpe computed on event-to-event marked-value increments ΔV_t as $\sqrt{n} \overline{\Delta V} / \text{sd}(\Delta V)$ without annualization, and behavioural quantities (mean absolute inventory, mean quoted spread, fills per episode).

Significance. Paired tests compare two agents on the same stock and seed before pooling differences across stocks. We report the mean paired difference, a 95% confidence interval, and the paired win rate over the 15 units, but treat the raw PnL magnitude as secondary to ND-PnL and the per-stock breakdown. Direct AS has no training seed, so we evaluate it on the same stock, day, episode-start, and replay-seed schedule and pair it with each of the five learned-policy seed indices. Because the units cluster in three stocks rather than 15 independent draws, we read the pooled test alongside the per-stock breakdown in Appendix D rather than treating the 15 as independent.

Table 1: Five-seed matched comparison. Values average 15 stock–seed units per method. PnL and Sharpe include standard errors across those units.

Method	PnL ($\pm$ SE)	ND-PnL	Sharpe ($\pm$ SE)	Abs. inv.	Spread	Fills
SOFT-AS PPO	75.17 $\pm$ 22.06	1223.9	2.54 $\pm$ 0.44	54.2	0.0468	511.7
PROFIT PPO	65.67 $\pm$ 22.51	1010.5	2.29 $\pm$ 0.42	26.6	0.0551	437.7
C-PPO	60.07 $\pm$ 21.38	812.1	2.07 $\pm$ 0.51	19.3	0.0637	333.1
Direct AS	26.64 $\pm$ 4.82	462.2	0.75 $\pm$ 0.12	157.5	0.0544	24.7

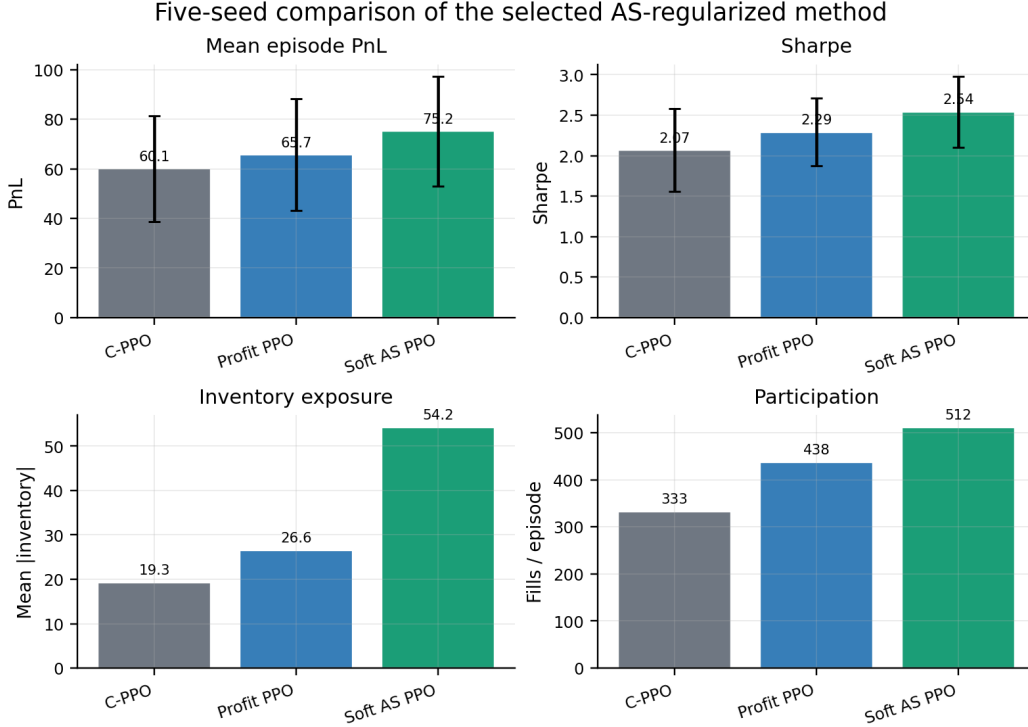


Figure 1: Headline performance and risk diagnostics over five seeds. SOFT-AS PPO has the strongest ND-PnL and Sharpe, but reaches them by quoting tighter and participating more, which raises inventory exposure.

6 Results

6.1 Soft AS improves return over both controls

Our primary comparison is the paired test together with ND-PnL, because the paired test holds the stock and seed fixed while ND-PnL removes most of the price-scale effect across stocks. Against the reward-shaped C-PPO baseline, SOFT-AS PPO improves per-unit PnL by 15.10 (95% CI [9.19, 21.01], $p = 8.1 \times 10^{-5}$) and wins on 14 of 15 units; against PROFIT PPO, which already removes the hybrid reward, it improves by 9.50 ($p = 1.9 \times 10^{-4}$) and again wins on 14 of 15 (Table 2). The second comparison is the more informative one: PROFIT PPO has stripped out the old reward entirely, so the gain over it is attributable to the calibrated AS prior rather than to dropping the hybrid terms.

The aggregate metrics in Table 1 tell the same story in scale-comparable form. On ND-PnL, which counts how many spreads each agent captures and is therefore comparable across stocks, SOFT-AS PPO reaches 1223.9 against 812.1 for C-PPO and 1010.5 for PROFIT PPO—roughly a 50% and 21% improvement—and it also leads on Sharpe (2.54 vs. 2.07 and 2.29). Raw mean PnL is reported

Table 2: Paired raw-PnL comparisons over 15 stock–seed units. The differences are computed within stock and seed; their scale should be read together with ND-PnL and the per-stock table.

Method	Baseline	Δ PnL	95% CI	p	Win rate
SOFT-AS PPO	C-PPO	15.10	[9.19, 21.01]	8.1×10^{-5}	0.93
PROFIT PPO	C-PPO	5.60	[0.36, 10.83]	0.038	0.73
SOFT-AS PPO	PROFIT PPO	9.50	[5.45, 13.56]	1.9×10^{-4}	0.93
SOFT-AS PPO	Direct AS	48.54	[11.42, 85.66]	0.014	0.80

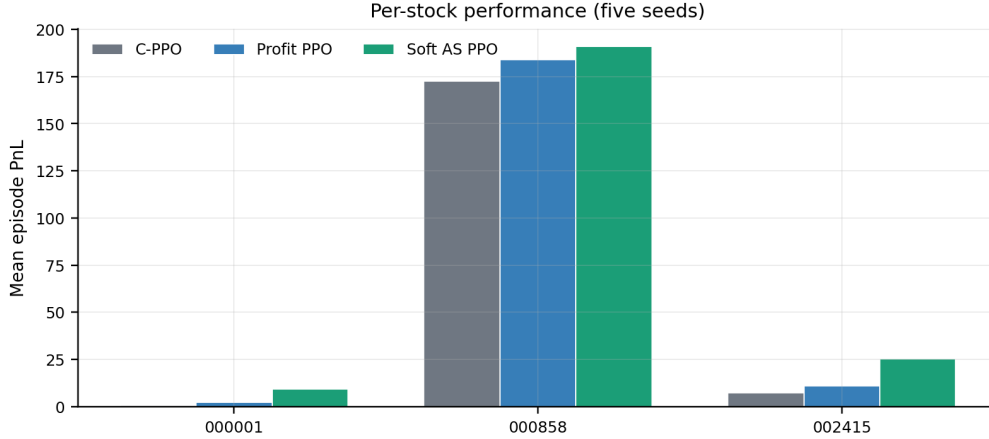


Figure 2: Per-stock mean episode PnL over five seeds. The proposed method improves clearly on the low-price and mid-price stocks and remains competitive on the high-price liquid stock.

for completeness but should be read with care: because the three stocks differ in price level, the mean PnL is dominated by the high-price stock and is not comparable across instruments.

That dominance does not, however, carry into the improvement. Figure 2 and Appendix D break the gain down by stock: SOFT-AS PPO improves ND-PnL and Sharpe over C-PPO on the low-price stock 000001 and the mid-price stock 002415, and remains competitive on the liquid high-price stock 000858, where the profit-only control is the strongest on ND-PnL. The gain is therefore spread across stocks rather than a single-stock artifact.

6.2 The policy departs from its teacher

Because SOFT-AS PPO is regularized toward AS, the natural question is whether it merely reproduces the teacher. It does not. Direct AS under the same low-risk per-stock calibration reaches 26.64 mean PnL, 462.2 ND-PnL, and 0.75 Sharpe (Table 1). SOFT-AS PPO improves on it by 48.54 PnL in the paired stock–seed comparison (95% CI [11.42, 85.66], $p = 0.014$) and wins 12 of the 15 units (Table 2). The result is not uniform across stocks: direct AS is competitive on the low-price stock 000001, while SOFT-AS PPO is much stronger on 000858 and 002415. The teacher-divergence diagnostics in Section 6.3 explain this pattern: the policy stays close to AS in action space, but the gap is not zero. The soft prior supplies structure without forcing the policy to copy the teacher everywhere.

6.3 The AS prior changes behaviour directly

The main advantage of AS regularization over reward shaping is inspectability: at evaluation time we can log both the policy action and the AS teacher action. Figure 3 shows that SOFT-AS PPO is substantially closer to the teacher in both quote-bias and spread coordinates. The mean L2 action gap falls from 0.468 for C-PPO and 0.469 for PROFIT PPO to 0.230 for SOFT-AS PPO.

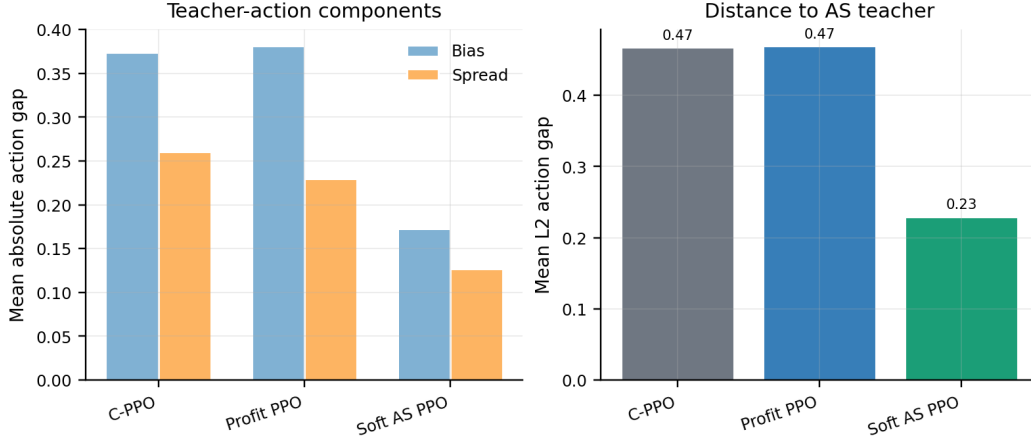


Figure 3: Teacher-divergence diagnostics from saved seed-0/1/2 checkpoints. SOFT-AS PPO remains much closer to the fitted AS teacher than either control, showing that the regularizer directly steers quoting behaviour rather than only changing terminal PnL.

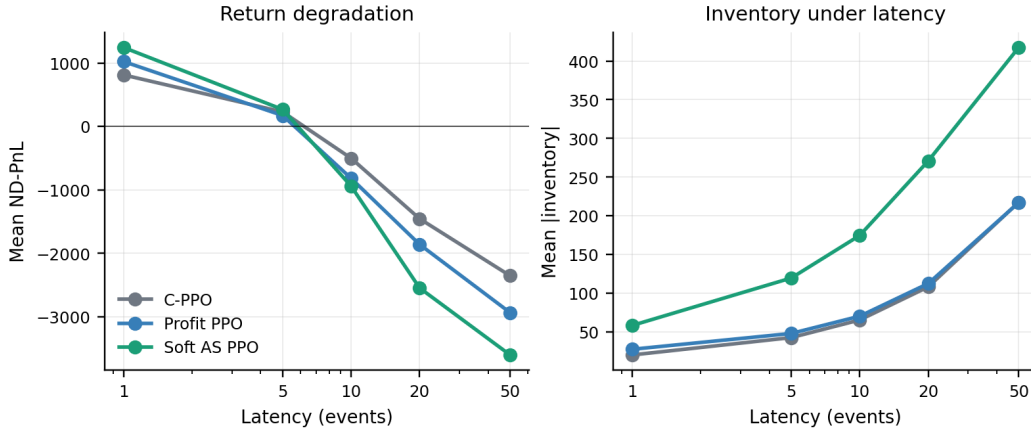


Figure 4: Latency sweep on saved seed-0/1/2 policies. SOFT-AS PPO is best at short latency but is more exposed at high latency because it participates more and carries more inventory.

Consistent with this, in the five-seed training evaluation SOFT-AS PPO has higher average inventory and more fills than both controls (Table 1). AS regularization improves the policy by making it more willing to quote, so the added return comes with a clearer inventory and latency tradeoff.

6.4 Latency remains the main weakness

To stress-test robustness we run an evaluation-only latency sweep on the saved checkpoints at 1, 5, 10, 20, and 50 events; the policies were trained at zero latency, so this measures how latency-naïve policies degrade rather than how a latency-aware agent would behave. At low latency SOFT-AS PPO has the best return. As latency grows all methods degrade, and SOFT-AS PPO degrades fastest (Figure 4): the same tight, high-participation quoting that wins at low latency becomes costly once quotes are stale and inventory is larger. This supports the low-latency result while marking exactly where an AS-regularized maker would need latency-aware or inventory-aware control.

6.5 Decision trace

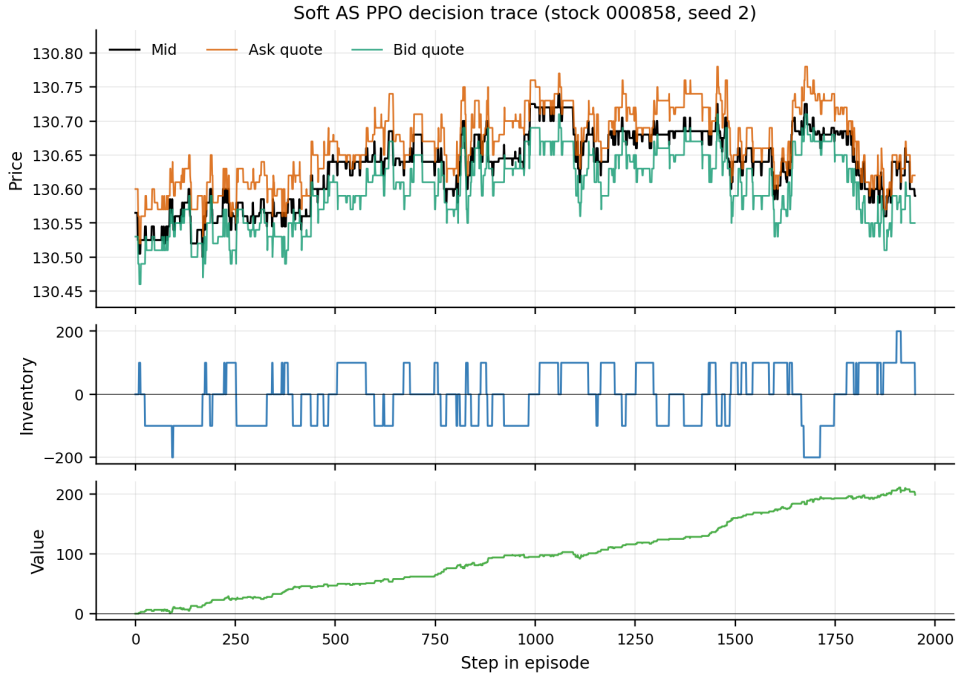


Figure 5: Decision trace for SOFT-AS PPO on stock 000858, seed 2. The policy quotes actively around the mid-price, manages inventory around zero, and accumulates positive marked-to-market value over the episode.

Figure 5 mirrors the original paper’s decision-trace diagnostic for the proposed method. It is descriptive rather than causal, but it shows that the improved policy still looks like a market maker (quoting both sides around mid and reverting inventory) rather than a sparse opportunistic trader.

7 Discussion

Why the soft prior helps. The three main methods separate the effect of the old reward from the effect of the AS prior. PROFIT PPO edges out C-PPO by a smaller and less certain margin (Table 2), suggesting the hybrid reward was at best no help in this synthetic setting. SOFT-AS PPO then beats PROFIT PPO more decisively, which points to the classical teacher adding useful structure beyond the reward change. Hard AS constraints, reported in Appendix B, were weaker because clipping the policy near AS removes profitable deviations. The soft prior works better because the policy can still move away from the teacher when the data support it. The latency results reinforce this point. Active quoting helps when information is fresh, but the same behavior becomes dangerous when quotes go stale. A rigid constraint or a shaped reward that discourages participation would completely hide this tradeoff. The soft prior makes it visible.

Scope and risk. The claim is intentionally narrow. On a paper-shaped synthetic benchmark, with per-stock policies and matched seeds, AS-regularized PPO is a stronger and more interpretable alternative to hybrid reward shaping. The method trades more and carries more inventory, and the latency sweep shows that this matters when quotes become stale. A live or proprietary-data claim would require a real event stream, transaction costs, market impact, and stricter tail-risk constraints. The synthetic data was also designed to make spread capture learnable, with only soft adverse selection. Real markets with heavier informed flow would likely be less kind to aggressive quoting strategies.

Practical value. Reward shaping hides design decisions inside scalar coefficients that interact indirectly. An AS action prior makes those decisions inspectable: the teacher can be calibrated, logged

at evaluation time, and stress-tested, and the policy’s distance from it can be measured directly (Section 6.3). When the teacher is too aggressive, γ , κ , or λ are economically meaningful knobs, which is easier to reason about than a hybrid reward whose terms trade off implicitly.

8 Conclusion

We revisited a reward-shaped deep market maker and replaced its hybrid reward with a pure-profit objective plus a soft Avellaneda–Stoikov action prior. The motivation was practical: hybrid rewards encode useful behavioural intentions but spreads them across coupled coefficients which are hard to interpret, to tune, and to debug when the agent misbehaves. An AS action prior replaces all of that with a single classical model whose parameters have clear economic meaning.

On a controlled synthetic replication of the original experimental setup, SOFT-AS PPO improves spread-normalized PnL by roughly 50% over reward-shaped C-PPO and by 21% over a profit-only control across five seeds per stock, and the gain is supported by paired tests and the per-stock breakdown. The Sharpe ratio also improves (2.54 against 2.07 for C-PPO), indicating that the higher returns are not simply noisier. Teacher-divergence diagnostics confirm that the regularizer directly steers quoting behaviour: on average, the policy stays close to the AS teacher but moves away from it when the data reward a different quote, which is the mechanism by which it outperforms the teacher and the unregularized baseline.

The improvement is not free. SOFT-AS PPO quotes tighter, fills more often, and carries approximately three times the inventory of C-PPO. The latency sweep shows that this active profile degrades faster when quotes become stale, which would matter in any deployment with non-negligible communication or computation delays. These are direct consequences of the method’s strength: a prior that encourages participation will always carry more inventory risk than one that discourages it.

Several questions remain open. A state-dependent or annealed λ could let the agent tighten its adherence to the teacher in volatile regimes and loosen it in stable ones. Most importantly, evaluation on a non-proprietary real LOB stream would test whether the AS prior remains useful when the fill process, adverse selection, and microstructure differ from the synthetic benchmark.

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Appendix

A Synthetic benchmark

The synthetic generator produces event-level LOB days with a latent fair-value process, bid–ask bounce, stable/trending/volatile regimes, market-order flow, and top-10 price/volume levels. Each day has 10,000 raw events before stable-window filtering; downstream evaluation keeps the original paper’s stable trading windows, 10:00–11:30 and 13:00–14:30. The three instruments use base prices 16.45 (000001), 130.00 (000858), and 35.00 (002415), giving low-, high-, and mid-price test cases with different realized spread and volatility scales.

The generator first simulates a latent fair value on the tick grid. Regime labels change the probability and size of fair-value moves, and order-flow intensity changes with the regime. Around the fair value, the code builds ask and bid ladders with ten levels of prices and volumes, then records trades whose signs and prices can cross the agent’s later quotes during replay. This gives the policy the same kind of 50×40 LOB tensor used by Guo et al. [8], while keeping the data fully reproducible and shareable. The replay simulator has no market impact and no transaction cost, so the benchmark should be read as controlled method-comparison evidence rather than market-deployment evidence.

B Teacher calibration and sensitivity

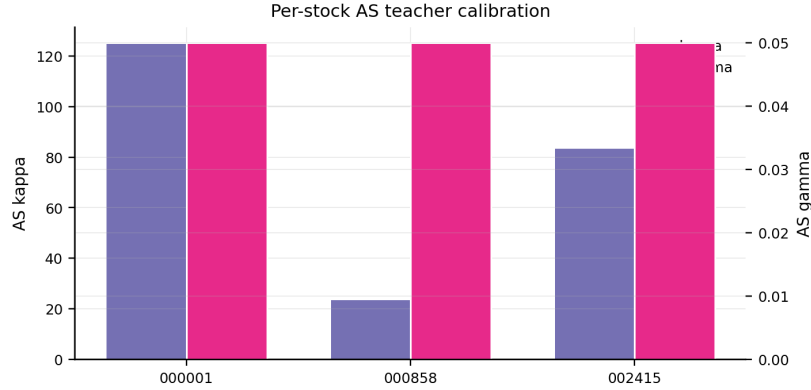


Figure 6: Per-stock AS teacher parameters used by the fitted direct-AS teacher and the AS-regularized policies.

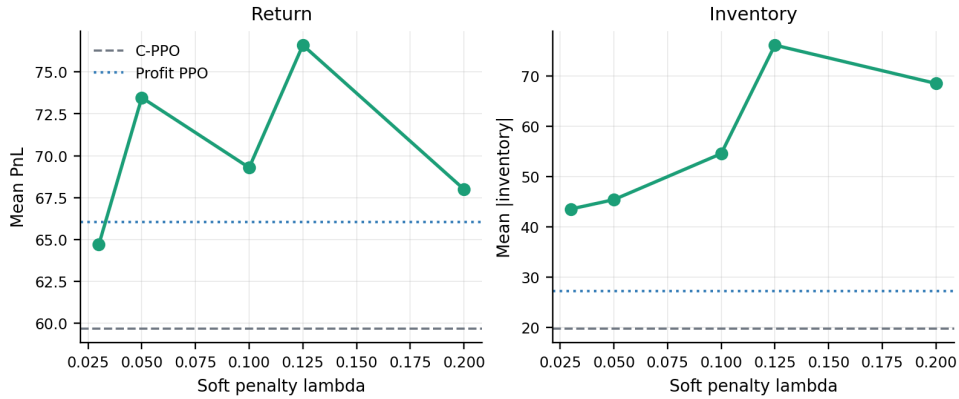


Figure 7: Sensitivity of soft AS regularization to λ in the original three-seed 400k sweep. The selected low-risk teacher is reported in the main paper; this plot shows that the base-teacher penalty weight has a broad useful region, while inventory generally rises with tighter quoting.

The broader matched sweep included base-teacher $\lambda \in \{0.03, 0.05, 0.10, 0.125, 0.20\}$, low-risk and high-risk teachers, fill- κ calibration, hard AS windows, direct AS, and BC warm start. The selected main method uses the low-risk teacher with $\lambda = 0.10$ because it gave the best balance of PnL, Sharpe, and behavioural alignment in the five-seed batch.

Hard AS applies the teacher as an action constraint rather than a training penalty: the PPO policy proposes an action, and the environment clips that action to a fixed window around the AS action before converting it to quotes. The wider hard window was the best hard-AS variant in the matched batch, but it still underperformed the selected soft regularizer. This is consistent with the role of the constraint. A hard window prevents the policy from taking large deviations even when the synthetic tape makes those deviations profitable. It also treats the AS teacher as locally correct at every step, which is too strong once the fill process, finite action bounds, and end-of-episode liquidation differ from the idealized AS assumptions. Hard AS is therefore useful as a guardrail diagnostic, but it is a less flexible way to combine the teacher with PPO.

BC warm start is different again. In the behavioural-cloning phase, the policy is trained by supervised learning to imitate AS actions on states from the training data; no RL reward is used in that phase. The resulting checkpoint is then fine-tuned with PPO using the pure-profit objective. This tests whether AS is best used only as an initialization. In exploratory runs it helped relative to training from scratch, but it was weaker than keeping the AS prior active during PPO. The likely reason is drift: once PPO fine-tuning begins, the policy can move away from AS without paying any continuing cost.

C Attention diagnostic

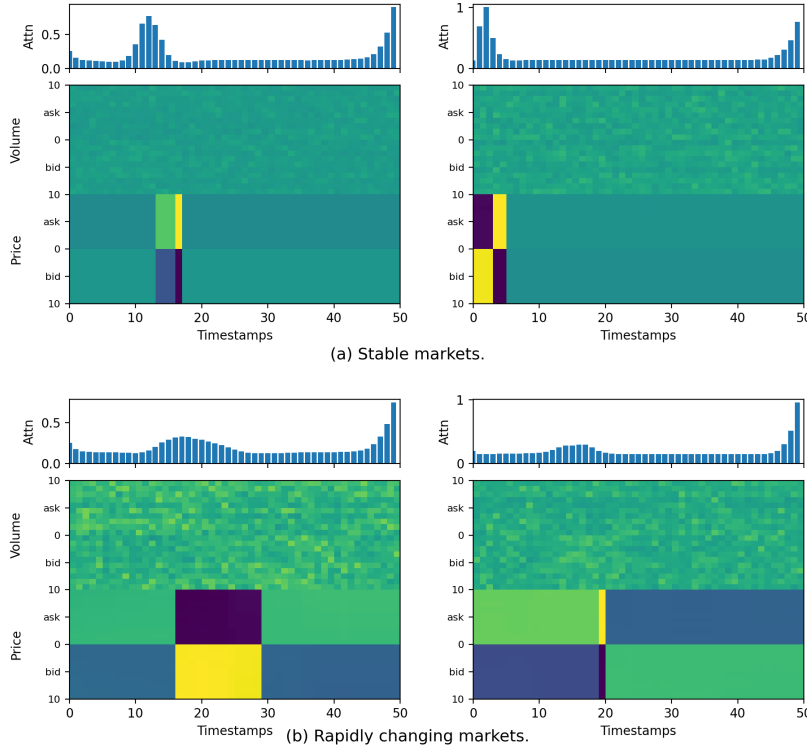


Figure 8: Attention diagnostic for the proposed policy. Low-variation and high-variation windows are selected by mid-price standard deviation over the 50-event input window. The plot is included for comparison with the original paper’s interpretability figure, but it should be read cautiously: attention maps are descriptive diagnostics, not causal explanations of trading performance.

The windows in Figure 8 are selected mechanically. For every possible 50-event input window, we compute the standard deviation of the level-1 mid-price. The two non-overlapping windows with the lowest values are labelled low-variation, and the two with the highest values are labelled high-variation. This makes the selection reproducible and matches the spirit of the original stable-versus-rapid comparison, but it is still a heuristic. We are confident that the selected windows differ in local mid-price variation under this metric; we should not claim that they are hand-validated economic regimes.

The upper panels show attention mass over the 50 input timestamps, aggregated from the Attn-LOB encoder’s self-attention weights. The lower panels show the normalized LOB tensor for the same window. These heatmaps can look unusual because the LOB tensor is not an image: it stacks ask/bid prices and volumes across ten levels, with normalization applied per window. Persistent price ladders and large volume bands therefore appear as blocks or stripes. Attention can also look fairly flat after aggregation because different heads attend to different timestamps, and summing or averaging over heads smooths those patterns. In this run the diagnostic mainly says that the encoder is using broad recent context rather than placing a sharp, paper-like spike on a single event. That is informative, but weaker than the attention story in the original paper and should be treated as a qualitative check rather than evidence of mechanism.

D Per-stock numerical results

Table 3: Per-stock means over five seeds.

Stock	Method	PnL	ND-PnL	Sharpe	Abs. inv.
000001	C-PPO	0.44	8.0	0.19	0.5
000001	PROFIT PPO	2.16	44.3	0.69	2.2
000001	SOFT-AS PPO	9.16	282.7	0.75	64.2
000001	Direct AS	10.05	363.4	0.31	214.8
000858	C-PPO	172.64	2292.5	4.66	52.1
000858	PROFIT PPO	183.83	2746.6	4.38	69.2
000858	SOFT-AS PPO	190.92	2689.4	4.67	62.4
000858	Direct AS	51.68	517.6	1.35	52.0
002415	C-PPO	7.15	135.6	1.36	5.1
002415	PROFIT PPO	11.02	240.5	1.81	8.3
002415	SOFT-AS PPO	25.44	699.4	2.20	36.2
002415	Direct AS	18.18	505.7	0.58	205.7